

A Hybrid Deep Reinforcement Learning and Agentic Framework for Real-Time Distillation Optimization

Integrating Rigorous DWSim Flowsheets with Hybrid DRL & Agentic framework

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ABSTRACT

Distillation remains the most energy-intensive operation in petroleum refining, where the optimization of multi-unit sequences is frequently challenged by non-linear thermodynamics and a lack of interpretability in automated control strategies. This study addresses the challenge of maximizing the economic margin between feedstock costs and product revenue—for an integrated distillation flowsheet comprising Atmospheric Distillation (ADU), Naphtha Stabilizer (NSU), and Vacuum Distillation (VDU) units. To navigate the 16-dimensional continuous action space and 120-dimensional state vector, three Deep Reinforcement Learning (DRL) architectures, Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3) were benchmarked within a custom Gymnasium environment interfaced with the DWSIM chemical process simulator. This optimization layer was augmented by a cognitive multi-agent framework powered by NVIDIA Nemotron 3 Nano 30B A3B, which utilized specialized personas to provide Explainable AI (XAI) narratives and safety-critical oversight of the RL-driven setpoint adjustments.

The experimental results indicated that PPO was the most effective architecture for this high-dimensional task, achieving a peak average reward of 22,592.33 while exhibiting the highest training stability and lowest computational latency. Inference trials confirmed that the trained agent successfully prioritized high-value fractions, such as Naphtha and Diesel, while reducing ASTM D95% quality specification violations by approximately 85% compared to initial training phases. Furthermore, the agent orchestrator successfully bridged the industrial trust gap by translating numerical control actions into validated engineering rationales and identifying potential overhead corrosion risks that were invisible to the numerical RL agents. This research provides a scalable blueprint for Cognitive Process Control, demonstrating that the synergy of autonomous Deep RL optimization and LLM-based reasoning can significantly enhance both the profitability and safety of complex refinery operations.

KEYWORDS

Deep Reinforcement Learning, Deep Q-Networks, Soft Actor Critic, Proximal Policy Optimization, TD3, DWSim, Optimization, AI Agent, Process Simulation

1 Introduction

1.1 Problem Statement

Crude Distillation Units (CDUs) are the foundational and most energy-intensive separation processes in petroleum refining. Crude Oil is processed in an oil refinery through multiple Unit Operations, to produce various finished products like LPG, Motor Spirit (Gasoline), Jet Fuel, Diesel, Low-Sulphur Fuel Oil, Bitumen and so on. The crude is first fractionated into multiple unfinished intermediates which are further refined into various finished products in subsequent processes. Traditionally, Advanced Process Control (APC) strategies, such as Model Predictive Control (MPC), have been utilized to maintain process variables within strict operational bounds. However, APC models are linear and bound by standalone strategies and lack ability to run holistic optimization. As refining margins become increasingly volatile, there is a critical need to transition from steady-state regulation to dynamic, market-driven overall optimization. While Deep Reinforcement Learning (DRL) has arisen as a great choice for solving highly non-linear, stochastic optimization problems in chemical processes (Spielberg et al., 2019), its industrial implementation remains severely hindered by the ‘black-box’ nature of neural networks. Operators and process engineers are inherently reluctant to deploy opaque control policies that lack physical interpretability (Venkatasubramanian, 2019). This study seeks to develop a solution which utilizes Deep Reinforcement Learning coupled with process Digital Twin, for economic and process optimization of Crude Distillation Unit.

1.2 Significance

This study bridges the critical gap between advanced artificial intelligence and practical

industrial operability. By maximizing thermodynamic efficiency and product yield in response to fluctuating market prices, refineries can realize substantial economic gains while minimizing energy waste and emissions. Additionally, introducing an LLM-based cognitive AI agent as an interpretability and reporting layer addresses the operator trust deficit. This ‘Explainable AI’ (XAI) approach ensures that human operators remain in the loop, capable of auditing and understanding the physicochemical rationale behind the AI's suggested setpoint changes (Nian et al., 2020).

1.3 Target End-Users

The primary end users of this hybrid AI framework are chemical plant operators, process engineers, and production planners. While the Deep RL agent performs high-dimensional mathematical optimization in the background, the LLM interface is designed specifically to communicate with these domain experts in natural language, translating thermodynamic state changes into actionable engineering reports.

1.4 Data Sources and Simulation Context

Reinforcement learning agents learn from experience while navigating through the internal environment. The training data for neural network to compute Q-value is generated from the simulation.

Various crude assay data is used for process simulation to ensure a robust model able to handle various types of feed quality (ExxonMobil, n.d.)

1.5 Final Deliverable

This study aims to successfully design, train, and deploy an AI-augmented digital twin for a CDU.

The application has a Django dashboard to interact with the backend system comprising of a process Digital Twin built on DWSim, an optimizer built on Deep Reinforcement Learning agent, and a LLM-based AI Agent which creates reports and interacts with the user for explainability. In the interface, the user will input current market prices, triggering the Deep RL agent to interact with the underlying DWSim physical model to discover optimal operating parameters. The final output is a set of optimized targets for the plant to operate and a comprehensive, natural-language engineering report generated by the LLM, detailing the economic trade-offs, thermodynamic constraints, and justification for the proposed optimization.

2 Data Summary

2.1 Input State Representation

Our study revolves around finding the most optimized manner of distillation of crude. Hence, one of the primary focuses in this study is to understand the properties of commonly available crude assays. The dataset chosen comprises physicochemical properties for five distinctive crude oil assays (Azeri Light, Erha, Tapis, Upper Zakum, and WTI Light) sourced from the ExxonMobil's open database (ExxonMobil, n.d.).

Whole Crude Physicochemical Attributes such as mass density, API gravity (the standard measure of extent of heaviness of oil), Total Sulfur content, and Total Acid Number (TAN) are critical defining factors of crude assays. These properties are critical as they define the intrinsic quality and potential corrosivity of the feed (González-Durán et al., 2024; Al-Sayegh et al., 2016). Concentrations of Nitrogen and metals such as Vanadium (V) and Nickel (Ni) are significant since they serve as

catalyst poisons in downstream units like fluid catalytic cracking (FCC) (Al-Sayegh et al., 2016). Rheological and Volatility Markers such as Kinematic Viscosity, Reid Vapor Pressure (RVP), and Pour Point characterize the flow behavior and thermal stability of the crude. The True Boiling Point (TBP) curve defines the volumetric and weight percentage yields for various product cuts (Naphtha, Kerosene, Diesel, and Residue). Quality metrics for these cuts—such as Octane Number (RON/MON) for Naphtha and Cetane Index for Diesel — are utilized as process constraints within downstream process optimizers (AlRijeb et al., 2025).

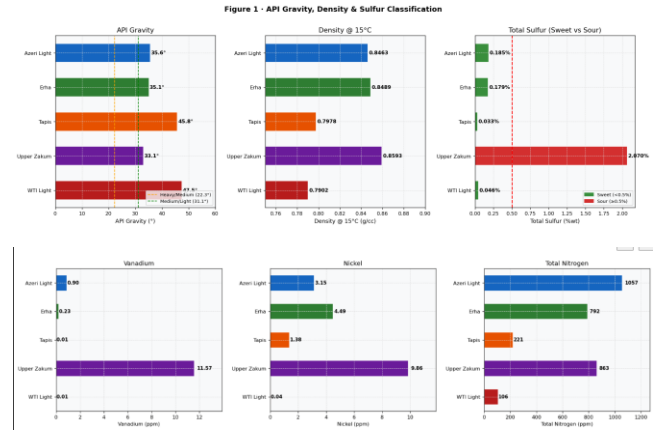


Fig 1. (a). API Gravity, Density (at 15 °C) and Total Sulphur distribution in the selected crude assays

Fig 1. (b) Metals and Total Nitrogen content in the selected crude assays

2.2 Fractional Yield Dynamics and TBP Analysis

Upon analysis, True Boiling Point (TBP) distillation curves derived from the dataset shed light on significant disparities in product slates. WTI Light and Tapis have demonstrated superior Naphtha yields, which are essential for gasoline production. Conversely, Erha and Tapis have showed a higher propensity for middle distillate

production like Kerosene and Diesel. Upper Zakum is characterized by a bottom-heavy profile, yielding a higher percentage of Vacuum Gas Oil (VGO) and atmospheric residues. Since the Deep RL agent is inherently aimed at driving margins, yield distributions directly impact the reward function of the optimizer; for instance, when market prices favor Jet Fuel, the optimizer must identify the specific operating temperatures within the digital twin that maximize the Kerosene cuts while maintaining thermodynamic stability (Li et al., 2025).

	Light Ends (C5-40°C)	Light Naphtha (40-100°C)	Heavy Naphtha (100-150°C)	Kerosene (150-200°C)	Jet Fuel (200-250°C)	Diesel (250-300°C)	15-400 (300-350°C)	Heavy 400 (350-400°C)	Heavy Residue (400°C+)	VGO (370-400°C)	MAO (400-450°C)	MAO (450-500°C)	MAO (500-550°C)	MAO (550-600°C)	MAO (600-650°C)	MAO (650-700°C)	MAO (700-750°C)	MAO (750-800°C)	MAO (800-850°C)	MAO (850-900°C)	MAO (900-950°C)	MAO (950-1000°C)	MAO (1000-1050°C)	MAO (1050-1100°C)	MAO (1100-1150°C)	MAO (1150-1200°C)	MAO (1200-1250°C)	MAO (1250-1300°C)	MAO (1300-1350°C)	MAO (1350-1400°C)	MAO (1400-1450°C)	MAO (1450-1500°C)	MAO (1500-1550°C)	MAO (1550-1600°C)	MAO (1600-1650°C)	MAO (1650-1700°C)	MAO (1700-1750°C)	MAO (1750-1800°C)	MAO (1800-1850°C)	MAO (1850-1900°C)	MAO (1900-1950°C)	MAO (1950-2000°C)	MAO (2000-2050°C)	MAO (2050-2100°C)	MAO (2100-2150°C)	MAO (2150-2200°C)	MAO (2200-2250°C)	MAO (2250-2300°C)	MAO (2300-2350°C)	MAO (2350-2400°C)	MAO (2400-2450°C)	MAO (2450-2500°C)	MAO (2500-2550°C)	MAO (2550-2600°C)	MAO (2600-2650°C)	MAO (2650-2700°C)	MAO (2700-2750°C)	MAO (2750-2800°C)	MAO (2800-2850°C)	MAO (2850-2900°C)	MAO (2900-2950°C)	MAO (2950-3000°C)	MAO (3000-3050°C)	MAO (3050-3100°C)	MAO (3100-3150°C)	MAO (3150-3200°C)	MAO (3200-3250°C)	MAO (3250-3300°C)	MAO (3300-3350°C)	MAO (3350-3400°C)	MAO (3400-3450°C)	MAO (3450-3500°C)	MAO (3500-3550°C)	MAO (3550-3600°C)	MAO (3600-3650°C)	MAO (3650-3700°C)	MAO (3700-3750°C)	MAO (3750-3800°C)	MAO (3800-3850°C)	MAO (3850-3900°C)	MAO (3900-3950°C)	MAO (3950-4000°C)	MAO (4000-4050°C)	MAO (4050-4100°C)	MAO (4100-4150°C)	MAO (4150-4200°C)	MAO (4200-4250°C)	MAO (4250-4300°C)	MAO (4300-4350°C)	MAO (4350-4400°C)	MAO (4400-4450°C)	MAO (4450-4500°C)	MAO (4500-4550°C)	MAO (4550-4600°C)	MAO (4600-4650°C)	MAO (4650-4700°C)	MAO (4700-4750°C)	MAO (4750-4800°C)	MAO (4800-4850°C)	MAO (4850-4900°C)	MAO (4900-4950°C)	MAO (4950-5000°C)	MAO (5000-5050°C)	MAO (5050-5100°C)	MAO (5100-5150°C)	MAO (5150-5200°C)	MAO (5200-5250°C)	MAO (5250-5300°C)	MAO (5300-5350°C)	MAO (5350-5400°C)	MAO (5400-5450°C)	MAO (5450-5500°C)	MAO (5500-5550°C)	MAO (5550-5600°C)	MAO (5600-5650°C)	MAO (5650-5700°C)	MAO (5700-5750°C)	MAO (5750-5800°C)	MAO (5800-5850°C)	MAO (5850-5900°C)	MAO (5900-5950°C)	MAO (5950-6000°C)	MAO (6000-6050°C)	MAO (6050-6100°C)	MAO (6100-6150°C)	MAO (6150-6200°C)	MAO (6200-6250°C)	MAO (6250-6300°C)	MAO (6300-6350°C)	MAO (6350-6400°C)	MAO (6400-6450°C)	MAO (6450-6500°C)	MAO (6500-6550°C)	MAO (6550-6600°C)	MAO (6600-6650°C)	MAO (6650-6700°C)	MAO (6700-6750°C)	MAO (6750-6800°C)	MAO (6800-6850°C)	MAO (6850-6900°C)	MAO (6900-6950°C)	MAO (6950-7000°C)	MAO (7000-7050°C)	MAO (7050-7100°C)	MAO (7100-7150°C)	MAO (7150-7200°C)	MAO (7200-7250°C)	MAO (7250-7300°C)	MAO (7300-7350°C)	MAO (7350-7400°C)	MAO (7400-7450°C)	MAO (7450-7500°C)	MAO (7500-7550°C)	MAO (7550-7600°C)	MAO (7600-7650°C)	MAO (7650-7700°C)	MAO (7700-7750°C)	MAO (7750-7800°C)	MAO (7800-7850°C)	MAO (7850-7900°C)	MAO (7900-7950°C)	MAO (7950-8000°C)	MAO (8000-8050°C)	MAO (8050-8100°C)	MAO (8100-8150°C)	MAO (8150-8200°C)	MAO (8200-8250°C)	MAO (8250-8300°C)	MAO (8300-8350°C)	MAO (8350-8400°C)	MAO (8400-8450°C)	MAO (8450-8500°C)	MAO (8500-8550°C)	MAO (8550-8600°C)	MAO (8600-8650°C)	MAO (8650-8700°C)	MAO (8700-8750°C)	MAO (8750-8800°C)	MAO (8800-8850°C)	MAO (8850-8900°C)	MAO (8900-8950°C)	MAO (8950-9000°C)	MAO (9000-9050°C)	MAO (9050-9100°C)	MAO (9100-9150°C)	MAO (9150-9200°C)	MAO (9200-9250°C)	MAO (9250-9300°C)	MAO (9300-9350°C)	MAO (9350-9400°C)	MAO (9400-9450°C)	MAO (9450-9500°C)	MAO (9500-9550°C)	MAO (9550-9600°C)	MAO (9600-9650°C)	MAO (9650-9700°C)	MAO (9700-9750°C)	MAO (9750-9800°C)	MAO (9800-9850°C)	MAO (9850-9900°C)	MAO (9900-9950°C)	MAO (9950-10000°C)	MAO (10000-10050°C)	MAO (10050-10100°C)	MAO (10100-10150°C)	MAO (10150-10200°C)	MAO (10200-10250°C)	MAO (10250-10300°C)	MAO (10300-10350°C)	MAO (10350-10400°C)	MAO (10400-10450°C)	MAO (10450-10500°C)	MAO (10500-10550°C)	MAO (10550-10600°C)	MAO (10600-10650°C)	MAO (10650-10700°C)	MAO (10700-10750°C)	MAO (10750-10800°C)	MAO (10800-10850°C)	MAO (10850-10900°C)	MAO (10900-10950°C)	MAO (10950-11000°C)	MAO (11000-11050°C)	MAO (11050-11100°C)	MAO (11100-11150°C)	MAO (11150-11200°C)	MAO (11200-11250°C)	MAO (11250-11300°C)	MAO (11300-11350°C)	MAO (11350-11400°C)	MAO (11400-11450°C)	MAO (11450-11500°C)	MAO (11500-11550°C)	MAO (11550-11600°C)	MAO (11600-11650°C)	MAO (11650-11700°C)	MAO (11700-11750°C)	MAO (11750-11800°C)	MAO (11800-11850°C)	MAO (11850-11900°C)	MAO (11900-11950°C)	MAO (11950-12000°C)	MAO (12000-12050°C)	MAO (12050-12100°C)	MAO (12100-12150°C)	MAO (12150-12200°C)	MAO (12200-12250°C)	MAO (12250-12300°C)	MAO (12300-12350°C)	MAO (12350-12400°C)	MAO (12400-12450°C)	MAO (12450-12500°C)	MAO (12500-12550°C)	MAO (12550-12600°C)	MAO (12600-12650°C)	MAO (12650-12700°C)	MAO (12700-12750°C)	MAO (12750-12800°C)	MAO (12800-12850°C)	MAO (12850-12900°C)	MAO (12900-12950°C)	MAO (12950-13000°C)	MAO (13000-13050°C)	MAO (13050-13100°C)	MAO (13100-13150°C)	MAO (13150-13200°C)	MAO (13200-13250°C)	MAO (13250-13300°C)	MAO (13300-13350°C)	MAO (13350-13400°C)	MAO (13400-13450°C)	MAO (13450-13500°C)	MAO (13500-13550°C)	MAO (13550-13600°C)	MAO (13600-13650°C)	MAO (13650-13700°C)	MAO (13700-13750°C)	MAO (13750-13800°C)	MAO (13800-13850°C)	MAO (13850-13900°C)	MAO (13900-13950°C)	MAO (13950-14000°C)	MAO (14000-14050°C)	MAO (14050-14100°C)	MAO (14100-14150°C)	MAO (14150-14200°C)	MAO (14200-14250°C)	MAO (14250-14300°C)	MAO (14300-14350°C)	MAO (14350-14400°C)	MAO (14400-14450°C)	MAO (14450-14500°C)	MAO (14500-14550°C)	MAO (14550-14600°C)	MAO (14600-14650°C)	MAO (14650-14700°C)	MAO (14700-14750°C)	MAO (14750-14800°C)	MAO (14800-14850°C)	MAO (14850-14900°C)	MAO (14900-14950°C)	MAO (14950-15000°C)	MAO (15000-15050°C)	MAO (15050-15100°C)	MAO (15100-15150°C)	MAO (15150-15200°C)	MAO (15200-15250°C)	MAO (15250-15300°C)	MAO (15300-15350°C)	MAO (15350-15400°C)	MAO (15400-15450°C)	MAO (15450-15500°C)	MAO (15500-15550°C)	MAO (15550-15600°C)	MAO (15600-15650°C)	MAO (15650-15700°C)	MAO (15700-15750°C)	MAO (15750-15800°C)	MAO (15800-15850°C)	MAO (15850-15900°C)	MAO (15900-15950°C)	MAO (15950-16000°C)	MAO (16000-16050°C)	MAO (16050-16100°C)	MAO (16100-16150°C)	MAO (16150-16200°C)	MAO (16200-16250°C)	MAO (16250-16300°C)	MAO (16300-16350°C)	MAO (16350-16400°C)	MAO (16400-16450°C)	MAO (16450-16500°C)	MAO (16500-16550°C)	MAO (16550-16600°C)	MAO (16600-16650°C)	MAO (16650-16700°C)	MAO (16700-16750°C)	MAO (16750-16800°C)	MAO (16800-16850°C)	MAO (16850-16900°C)	MAO (16900-16950°C)	MAO (16950-17000°C)	MAO (17000-17050°C)	MAO (17050-17100°C)	MAO (17100-17150°C)	MAO (17150-17200°C)	MAO (17200-17250°C)	MAO (17250-17300°C)	MAO (17300-17350°C)	MAO (17350-17400°C)	MAO (17400-17450°C)	MAO (17450-17500°C)	MAO (17500-17550°C)	MAO (17550-17600°C)	MAO (17600-17650°C)	MAO (17650-17700°C)	MAO (17700-17750°C)	MAO (17750-17800°C)	MAO (17800-17850°C)	MAO (17850-17900°C)	MAO (17900-17950°C)	MAO (17950-18000°C)	MAO (18000-18050°C)	MAO (18050-18100°C)	MAO (18100-18150°C)	MAO (18150-18200°C)	MAO (18200-18250°C)	MAO (18250-18300°C)	MAO (18300-18350°C)	MAO (18350-18400°C)	MAO (18400-18450°C)	MAO (18450-18500°C)	MAO (18500-18550°C)	MAO (18550-18600°C)	MAO (18600-18650°C)	MAO (18650-18700°C)	MAO (18700-18750°C)	MAO (18750-18800°C)	MAO (18800-18850°C)	MAO (18850-18900°C)	MAO (18900-18950°C)	MAO (18950-19000°C)	MAO (19000-19050°C)	MAO (19050-19100°C)	MAO (19100-19150°C)	MAO (19150-19200°C)	MAO (19200-19250°C)	MAO (19250-19300°C)	MAO (19300-19350°C)	MAO (19350-19400°C)	MAO (19400-19450°C)	MAO (19450-19500°C)	MAO (19500-19550°C)	MAO (19550-19600°C)	MAO (19600-19650°C)	MAO (19650-19700°C)	MAO (19700-19750°C)	MAO (19750-19800°C)	MAO (19800-19850°C)	MAO (19850-19900°C)	MAO (19900-19950°C)	MAO (19950-20000°C)	MAO (20000-20050°C)	MAO (20050-20100°C)	MAO (20100-20150°C)	MAO (20150-20200°C)	MAO (20200-20250°C)	MAO (20250-20300°C)	MAO (20300-20350°C)	MAO (20350-20400°C)	MAO (20400-20450°C)	MAO (20450-20500°C)	MAO (20500-20550°C)	MAO (20550-20600°C)	MAO (20600-20650°C)	MAO (20650-20700°C)	MAO (20700-20750°C)	MAO (20750-20800°C)	MAO (20800-20850°C)	MAO (20850-20900°C)	MAO (20900-20950°C)	MAO (20950-21000°C)	MAO (21000-21050°C)	MAO (21050-21100°C)	MAO (21100-21150°C)	MAO (21150-21200°C)	MAO (21200-21250°C)	MAO (21250-21300°C)	MAO (21300-21350°C)	MAO (21350-21400°C)	MAO (21400-21450°C)	MAO (21450-21500°C)	MAO (21500-21550°C)	MAO (21550-21600°C)	MAO (21600-21650°C)	MAO (21650-21700°C)	MAO (21700-21750°C)	MAO (21750-21800°C)	MAO (21800-21850°C)	MAO (21850-21900°C)	MAO (21900-21950°C)	MAO (21950-22000°C)	MAO (22000-22050°C)	MAO (22050-22100°C)	MAO (22100-22150°C)	MAO (22150-22200°C)	MAO (22200-22250°C)	MAO (22250-22300°C)	MAO (22300-22350°C)	MAO (22350-22400°C)	MAO (22400-22450°C)	MAO (22450-22500°C)	MAO (22500-22550°C)	MAO (22550-22600°C)	MAO (22600-22650°C)	MAO (22650-22700°C)	MAO (22700-22750°C)	MAO (22750-22800°C)	MAO (22800-22850°C)	MAO (22850-22900°C)	MAO (22900-22950°C)	MAO (22950-23000°C)	MAO (23000-23050°C)	MAO (23050-23100°C)	MAO (23100-23150°C)	MAO (23150-23200°C)	MAO (23200-23250°C)	MAO (23250-23300°C)	MAO (23300-23350°C)	MAO (23350-23400°C)	MAO (23400-23450°C)	MAO (23450-23500°C)	MAO (23500-23550°C)	MAO (23550-23600°C)	MAO (23600-23650°C)	MAO (23650-23700°C)	MAO (23700-23750°C)	MAO (23750-23800°C)	MAO (23800-23850°C)	MAO (23850-23900°C)	MAO (23900-23950°C)	MAO (23950-24000°C)	MAO (24000-24050°C)	MAO (24050-24100°C)	MAO (24100-24150°C)	MAO (24150-24200°C)	MAO (24200-24250°C)	MAO (24250-24300°C)	MAO (24300-24350°C)	MAO (24350-24400°C)	MAO (24400-24450°C)	MAO (24450-24500°C)	MAO (24500-24550°C)	MAO (24550-24600°C)	MAO (24600-24650°C)	MAO (24650-24700°C)	MAO (24700-24750°C)	MAO (24750-24800°C)	MAO (24800-24850°C)	MAO (24850-24900°C)	MAO (24900-24950°C)	MAO (24950-25000°C)	MAO (25000-25050°C)	MAO (25050-25100°C)	MAO (25100-25150°C)	MAO (25150-25200°C)	MAO (25200-25250°C)	MAO (25250-25300°C)	MAO (25300-25350°C)	MAO (25350-25400°C)	MAO (25400-25450°C)	MAO (25450-25500°C)	MAO (25500-25550°C)	MAO (25550-25600°C)	MAO (25600-25650°C)	MAO (25650-25700°C)	MAO (25700-25750°C)	MAO (25750-25800°C)	MAO (25800-25850°C)	MAO (25850-25900°C)	MAO (25900-25950°C)	MAO (25950-26000°C)	MAO (26000-26050°C)	MAO (26050-26100°C)	MAO (26100-26150°C)	MAO (26150-26200°C)	MAO (26200-26250°C)	MAO (26250-26300°C)	MAO (26300-26350°C)	MAO (26350-26400°C)	MAO (26400-26450°C)	MAO (26450-26500°C)	MAO (26500-26550°C)	MAO (26550-26600°C)	MAO (26600-26650°C)	MAO (26650-26700°C)	MAO (26700-26750°C)	MAO (26750-26800°C)	MAO (26800-26850°C)	MAO (26850-26900°C)	MAO (26900-26950°C)	MAO (26950-27000°C)	MAO (27000-27050°C)	MAO (27050-27100°C)	MAO (27100-27150°C)	MAO (27150-27200°C)	MAO (27200-27250°C)	MAO (27250-27300°C)	MAO (27300-27350°C)	MAO (27350-27400°C)	MAO (27400-27450°C)	MAO (27450-27500°C)	MAO (27500-27550°C)	MAO (27550-27600°C)	MAO (27600-27650°C)	MAO (27650-27700°C)	MAO (27700-27750°C)	MAO (27750-27800°C)	MAO (27800-27850°C)	MAO (27850-27900°C)	MAO (27900-27950°C)	MAO (27950-28000°C)	MAO (28000-28050°C)	MAO (28050-28100°C)	MAO (28100-28150°C)	MAO (28150-28200°C)	MAO (28200-28250°C)	MAO (28250-28300°C)	MAO (28300-28350°C)	MAO (28350-28400°C)	MAO (28400-28450°C)	MAO (28450-28500°C)	MAO (28500-28550°C)	MAO (28550-28600°C)	MAO (28600-28650°C)	MAO (28650-28700°C)	MAO (28700-28750°C)	MAO (28750-28800°C)	MAO (28800-28850°C)	MAO (28850-28900°C)	MAO (28900-28950°C)	MAO (28950-29000°C)	MAO (29000-29050°C)	MAO (29050-29100°C)	MAO (29100-29150°C)	MAO (29150-29200°C)	MAO (29200-29250°C)	MAO (29250-29300°C)	MAO (29300-29350°C)	MAO (29350-29400°C)	MAO (29400-29450°C)	MAO (29450-29500°C)	MAO (29500-29550°C)	MAO (29550-29600°C)	MAO (29600-29650°C)	MAO (29650-29700°C)	MAO (29700-29750°C)	MAO (29750-29800°C)	MAO (29800-29850°C)	MAO (29850-29900°C)	MAO (29900-29950°C)	MAO (29950-30000°C)	MAO (30000-30050°C)	MAO (30050-30100°C)	MAO (30100-30150°C)	MAO (30150-30200°C)	MAO (30200-30250°C)	MAO (30250-30300°C)	MAO (30300-30350°C)	MAO (30350-30400°C)	MAO (30400-30450°C)	MAO (30450-30500°C)	MAO (30500-30550°C)	MAO (30550-30600°C)	MAO (30600-30650°C)	MAO (30650-30700°C)	MAO (30700-30750°C)	MAO (30750-30800°C)	MAO (30800-30850°C)	MAO (30850-30900°C)	MAO (30900-30950°C)	MAO (30950-31000°C)	MAO (31000-31050°C)	MAO (31050-31100°C)	MAO (31100-31150°C)	MAO (31150-31200°C)	MAO (31200-31250°C)	MAO (31250-31300°C)	MAO (31300-31350°C)	MAO (31350-31400°C)	MAO (31400-31450°C)	
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Crude	Classification	Origin	API	Refiner (Barrel)	CDU/CDU Processability	Refiner Feed Quality	CC Feed Quality	Hydrocracker Demand	Jet Fuel Potential	Direct Crude Quality	Petrochemical Feed	Crack/Feed Challenges
Front Light	Light Sweet	American	35.0	0.185	5	4	3	2	4	4	4	3
End	Light Sweet	Nigerian	35.1	0.178	5	4	3	3	4	5	4	2
Top	Light Ultra Sweet	Malaysian	40.0	0.003	5	4	4	5	5	5	4	4
Upper Stream	Light Sour	UAE	33.1	2.070	4	3	2	1	3	3	3	1
WT Light	Light Ultra Sweet	Iranian UDC	47.5	0.040	5	4	5	5	5	5	4	5

Fig 4. Crude-wise comparison based on its likely process and economic impact on a typical refinery as a whole

3 Literature Review

The traditional hierarchy of refinery control inherently resembles a temporal decoupling between the Real-Time Optimization (RTO) and Advanced Process Control (APC) layers, where the RTO typically samples unit states and recalculates economic targets, subject to model convergence, at a set interval of one to two hours. This multi-hour sampling gap, combined with the significant computational latency inherent in mathematical optimizers reaching convergence for complex non-linear models, forces the APC layer to manage transient process behavior and disturbances without updated economic guidance, often resulting in sustained periods of suboptimal operation. Recent advancements in Dynamic Real-Time Optimization (DRTO) attempt to bridge this gap. However, the computational cost of solving a rigorous, first-principles model in real-time still remains. This is where the integration of simulation model with a Deep RL agent offers a superior alternative. During training phase, by treating the simulation model as a high-fidelity simulator, the agent learns a control policy $\pi_{\theta}(a|s)$ that directly maps plant observations to optimal setpoints. Unlike the Sequential Quadratic Programming (SQP) methods described by Biegler (2010), which require a derivative-based search through the variable space at every time step, the Deep RL agent performs an instantaneous inference, significantly reducing the latency.

The predominant paradigm in industrial Real-Time Optimization (RTO) has historically relied on Non-Linear Programming (NLP) and Successive

Quadratic Programming (SQP) to solve steady-state models. While mathematically rigorous, these ‘equation-oriented’ solvers are often criticized for their ‘rigidity and susceptibility to convergence failures’ when faced with the high dimensionality of Crude Distillation Units (CDUs) (Biegler, 2010). A recurring deficiency in these mathematical optimizers is their inability to capture transient dynamics without prohibitive computational overhead (Darby et al., 2011).

A persistent challenge in mathematical RTO is the trade-off between profitability and the ‘Safe Operating Envelope’. Mathematical optimizers often operate at the ‘active constraints’ (Darby et al., 2011), leaving little room for error during stochastic disturbances. The shift toward Safe Reinforcement Learning, as highlighted by the emergence of the Constrained Policy Optimization (CPO) and Lagrangian RL frameworks (Achiam J., 2017), represents a significant trend in recent academic literature.

The integration of DWSIM—a CAPE-OPEN compliant, first-principles simulator—serves as a critical corrective. By utilizing DWSIM as the ‘ground truth’ environment, the DRL agent is not merely performing statistical curve fitting; it is interacting with a thermodynamically consistent system governed by Equations of State. This ‘First-Principle Compliance’ ensures that the learned control policy $\pi(a|s)$ remains bounded by physical reality, effectively bridging the gap between pure data-driven machine learning and rigorous chemical engineering (Spielberg et al., 2019).

In the context of a CDU, the state space S is characterized by a high-dimensional vector containing stage temperatures T_i , pressures P_i , mass fractions of pseudocomponents across the distillation curve, and product yields.

A critical contribution by Wang et al. (2024) in the Journal of Process Systems Engineering highlights the importance of Feature Engineering in RL for refineries. It is argued that normalizing states relative to the Feed Assay (the specific TBP curve of the incoming crude) allows the agent to generalize its learning across different crude blends—a concept very similar to well-known Transfer Learning where a model’s weight from learned from a larger dataset is partially fine-tuned for a specific domain use-case.

The reward function R_t is the most critical design element for any optimizer, which may be formulated as:

$$R_t = \sum_{p \in \text{Products}} (m_p \cdot V_p) - \sum_{u \in \text{Utilities}} (m_u \cdot C_u) - \sum_{c \in \text{Constraints}} \lambda_c \cdot \max(0, g_c(s) - \text{limit}_c)$$

Here, V_p represents the market value of product p , C_u is the cost of utilities (steam and cooling water), and λ_c is a penalty coefficient for violating quality constraints like Flash Point or ASTM D86 95% points. Literature review of recent times shows use of Lagrangian multipliers within the RL framework to dynamically adjust these penalties, ensuring that the agent prioritizes safety and product specifications over raw yield during periods of high process volatility (Stooke A., et al., 2020).

Academic research on Constrained Policy Optimization (CPO) algorithm shows that it is used to ensure that the agent remains within the Safe Operating Envelope of a VDU-ADU sequence. Their findings suggest that while CPO may converge slightly slower than standard PPO, it effectively eliminates the risk of dry trays or column flooding during the learning process (Achiam L., et al., 2017).

While the RL agent maximizes the profit-based reward function R_t , it often lacks the semantic awareness to identify nuanced process risks, such as dew point corrosion. Traditional RTOs focus on economic optima and may push the column toward temperatures that inadvertently lead to the condensation of acidic water in the overhead system. Recent studies in 2025 have proposed the use of Large Language Models (LLMs) as ‘Cognitive AI Agents’ that sit atop the optimization layer (Bran A. M., et al., 2024).

4 Methodology

4.1. Process Simulation

The process simulation is developed using DWSIM, a CAPE-OPEN compliant flowsheet simulator, utilizing a comprehensive model that encompasses an Atmospheric Distillation Unit (ADU), a Naphtha Stabilizer Unit (NSU), and a Vacuum Distillation Unit (VDU). This multi-unit sequence is configured to generate 13 distinct product streams, ranging from light gases to heavy residues. The ADU section produces five streams including un-condensed gas, Unstabilized Naphtha, Heavy Naphtha (HN), Straight-Run Kerosene Oil (SKO), Light Gas oil (LGO), and Heavy Gas oil (HGO). The NSU further refines light components of Unstabilized Naphtha into stabilizer off-gas, LPG, and stabilized run naphtha (SRN), while the VDU yields vacuum off-gas, vacuum diesel, Vacuum gas oil (VGO), Hotwell oil, and vacuum residue, while feeding from bottoms of ADU which is referred to as Reduced Crude Oil (RCO). For academic purpose, the flowsheet was converged on Erha Crude assay. The flowsheets developed are shown in Fig 4.

The crude feed is defined at ambient temperature and atmospheric pressure which is then pressurized through booster pumps, and heated in two phases

upto 360 °C where 56% of the crude turns in vapour phase.

The Atmospheric Distillation Unit (ADU) is configured with 30 stages, at 1.7 bar Top pressure with 0.4 bar pressure drop. Four side drawn products are defined. A partial condenser with initial reflux ratio of 5 used. Bottom temperature is set at 365 °C. The overhead liquid stream, Unstabilized Naphtha is drawn and fed to Naphtha Stabilizer to separate LPG from Straight-run Naphtha. The bottom, Reduced Crude Oil (RCO), is fed to Vacuum Distillation Unit for further refinement, after heating upto 405 °C which vaporizes 52% of the feed.

Naphtha Stabilizer (NSU) is configured with 30 stages, at 9.5 bar with 0.4 bar pressure drop, a partial condenser with 0.05 bar pressure drop, reflux ratio of 5 and bottom temperature at 155 °C.

Vacuum Distillation Unit (VDU) is configured with 20 stages. The top pressure is set at 0.2 bar to resemble a near-vacuum environment, while the column pressure drop is configured at 0.5 bar. A partial condenser is configured with a pressure drop of 0.1 bar and a reflux ratio of 5. The bottom temperature is set at 360 °C.

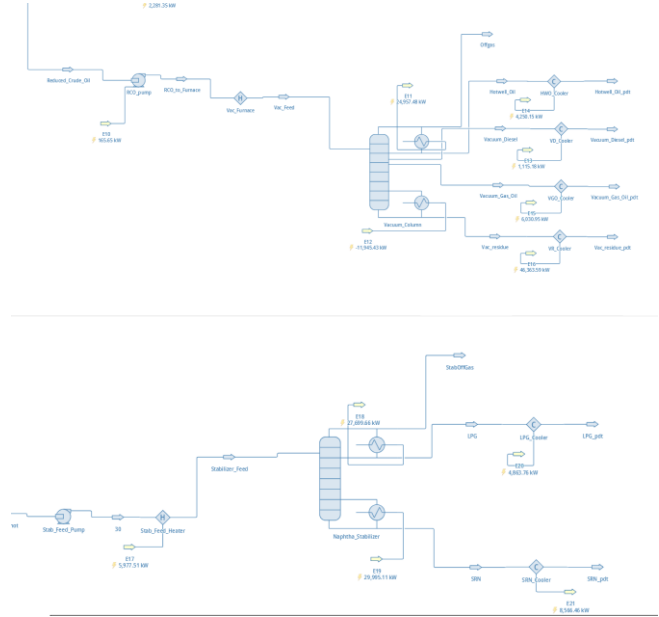
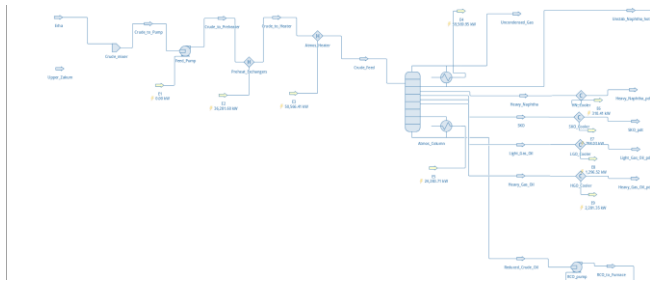


Fig 4. Flowsheets created depicting a typical Crude Distillation Unit

4.2. Reinforcement Learning Architectures

4.2.1. Justification for RL-based Optimization

The primary driver for implementing a Reinforcement Learning (RL) agent in a distillation context—specifically for the ADU, NSU, and VDU units—is the high sensitivity of process economics to volatile market conditions. Traditional Advanced Process Control (APC) and Model Predictive Control (MPC) systems often focus on maintaining setpoints or optimizing within a fixed linear regime. However, distillation units are the most energy-intensive components of a refinery, and their profitability is defined by the Crude Spread—the difference between the variable cost of feedstocks and the market value of products like Naphtha, Gasoil, and Residue (Sutton & Barto, 2018). An RL agent is uniquely capable of dynamic optimization because it can incorporate a reward function that explicitly represents the instantaneous economic potential:

$$R_t = \sum_{p \in \text{Products}} (m_p \cdot V_p) - \sum_{u \in \text{Utilities}} (m_u \cdot C_u) - \sum_{c \in \text{Constraints}} \lambda_c \cdot \max(0, g_c(s) - \text{limit}_c)$$

By training in a high-fidelity environment like DWSIM, the agent learns to shift cut points and adjust reflux ratios in real-time as market prices fluctuate, capturing lost profit that static controllers miss during transition states or shifts in crude composition (Shin et al., 2019).

4.2.2. Problem Formulation

The optimization of a distillation sequence is formulated as a Model-Free Reinforcement Learning problem, specifically categorized as a Continuous State-Action Markov Decision Process (MDP). In this framework, the state space S is defined by high-dimensional process variables, including stage temperatures, pressures, and molar fractions across the 40-stage columns. The action space A is continuous, consisting of setpoint adjustments for reflux ratios, reboiler duties, and pump-around flow rates. This problem is particularly challenging due to the non-linear dynamics and the delayed feedback inherent in chemical processes; an action taken at time t may not manifest its full effect on product purity until $t + n$. Because the underlying transition dynamics $P(s'|s, a)$ of a CDU are governed by complex thermodynamics (such as the Peng-Robinson or NRTL models), they are difficult to model analytically for control. Thus, model-free RL is employed which learns the optimal policy $\pi(a|s)$ through direct interaction with the simulator (Hubbs et al., 2020).

4.2.3. Candidate Algorithms and Selection Rationale

For this study, three state-of-the-art Deep RL architectures were selected for benchmarking: Proximal Policy Optimization (PPO), Soft Actor-

Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3). These were chosen because they are specifically designed to handle the curse of dimensionality and the continuous nature of chemical process control (Hubbs et al., 2020).

1. Proximal Policy Optimization (PPO): PPO was selected for its industry-standard stability and ease of tuning. As an on-policy algorithm, PPO ensures that policy updates do not deviate too far from the previous iteration by using a clipped objective function. This prevents the collapse of the simulation—a common issue in DWSIM when radical setpoint changes lead to non-convergence. PPO strikes a balance between ease of implementation and robust performance in high-dimensional spaces (Schulman et al., 2017).

2. Soft Actor-Critic (SAC): SAC was chosen for its superior sample efficiency and exploration capabilities. Unlike deterministic algorithms, SAC utilizes an "entropy-regularized" framework where the agent is rewarded not only for maximizing profit but also for acting as randomly as possible (maximizing entropy). In the context of distillation, this encourages the agent to discover unconventional operating windows that might be more profitable but are far from the standard operating procedures (Haarnoja et al., 2018).

3. Twin Delayed Deep Deterministic Policy Gradient (TD3): TD3 was included to address the systematic overestimation of value functions (Q-values) common in earlier algorithms like DDPG. By using two critic networks and delaying policy updates, TD3 provides a more conservative and accurate estimate of the long-term economic value of a process state. This is critical in refinery operations where overestimating the benefit of a setpoint change could lead to off-spec products and significant financial loss (Fujimoto et al., 2018).

4.2.4. Technical Mechanics of Candidate Algorithms

Proximal Policy Optimization (PPO) Mechanics:

PPO operates on the principle of a Trust Region, but simplifies the computation through a surrogate loss function. It calculates the probability ratio between the new policy and the old policy, $r_t(\theta)$. To ensure the update is incremental, it clips this ratio within a range (typically $[0.8, 1.2]$). Mathematically, the objective is $L^{CLIP}(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)]$, where A_t is the estimated advantage. This mechanism ensures that the distillation agent does not make catastrophic changes to the reflux or heat duties based on a single lucky episode (Schulman et al., 2017).

Soft Actor-Critic (SAC) Mechanics:

SAC is an off-policy, actor-critic algorithm that incorporates an alpha (α) parameter to control the temperature of the entropy term. The value function includes an entropy term $H(\pi(\cdot|s_t))$, leading to the objective $J(\pi) = \sum_{t=0}^T E_{(s_t, a_t) \sim \rho_\pi}[r(s_t, a_t) + \alpha H(\pi(\cdot|s_t))]$. In the Crude Distillation Unit environment, this allows the agent to maintain a diverse set of strategies for different crude types, as the temperature α automatically adjusts to encourage exploration when the agent is uncertain about the process response (Haarnoja et al., 2018).

Twin Delayed DDPG (TD3) Mechanics:

TD3 improves upon the DDPG baseline through three key innovations: Clipped Double-Q Learning, Target Policy Smoothing, and Delayed Policy Updates. It maintains two independent critic networks and uses the minimum value between them to calculate the target, which suppresses the overestimation of the profitability of certain process states. By updating the actor network less frequently compared to the critic network, it allows

the value estimation to stabilize before the policy changes, which is vital for handling the slow-moving thermal dynamics of large distillation columns (Fujimoto et al., 2018).

4.3. DWSim-RL Bridge

A thin, thread-safe wrapper is implemented around the DWSIM Automation3 COM/.NET API that is designed to isolate the rest of the codebase from Common Language Runtime (CLR) details and provides high-level, unit-consistent read/write operations for the flowsheet. The module established communication between the DWSim flowsheet and RL agent. The module begins with bootstrap utilities and constant mappings (object and stream names, product lists, property-code comments) and then defines a DWSIMBridge class that contains lifecycle control, safe reflection access, deterministic unit conversions, product/column readouts, D95 estimation from composition, controlled write/apply semantics for incremental actions, and disturbance injection. Logging is performed with loguru and the implementation favors defensive error handling.

The CLR bootstrap function ensures the required DWSIM assemblies are added to the Python process via the Python for .NET clr module. It uses a module-level lock and a boolean flag to make the operation idempotent and thread-safe; repeated calls are no-ops after the first successful bootstrap. This prevents multiple threads from attempting to load CLR assemblies concurrently and records status via logger messages.

4.4. Agent Training

The training process is centered on a high-fidelity interaction loop between the Reinforcement Learning agents and a custom-developed Gymnasium environment. The custom gymnasium environment serves as a wrapper for the DWSIM chemical process simulator, enabling the agent to manipulate the parameters associated with the flowsheet. To facilitate robust learning, the environment translates the physical constraints of the distillation process into a Markov Decision Process (MDP). Such integration of high-fidelity simulators with RL frameworks is critical to ensure that the agent learns policies grounded in rigorous thermodynamics rather than simplified linear approximations (Hubbs et al., 2020). The training was executed using the Stable Baselines3 library, a reliable framework for implementing state-of-the-art Deep RL algorithms with standardized hyperparameter configurations (Raffin et al., 2021).

The defined state space consists of a 120-dimensional observation vector. This resembles a comprehensive snapshot of the columns’ internal dynamics, including stage-wise temperatures, molar fractions, and product flow rates. This high-dimensional input allows the agent to perceive the non-linear sensitivities inherent in multi-unit distillation sequences. Conversely, the action space is defined as a 10-dimensional continuous vector, representing critical setpoints such as reflux ratios, and reboiler duties. Managing such a continuous action space is mathematically demanding, requiring algorithms capable of efficient exploration without violating the operational stability of the simulation (Sutton & Barto, 2018).

4.5. Reward Engineering

A major component in RL agent design is the reward function, which aims to align the agent’s

objectives with refinery profitability. The reward is calculated based on an economic objective function:

$$R = \Sigma(\text{Revenue}_{\text{products}}) - \Sigma(\text{Costs}_{\text{feed}} + \text{Costs}_{\text{utilities}}).$$

This formulation ensures that the agent prioritizes high-value product yields, such as Naphtha and Diesel, while minimizing the energy intensity of the reboilers and heaters. To stabilize the training of the critic networks, the raw economic reward is scaled by a factor of 10^{-3} , preventing gradient explosion during the early stages of backpropagation (Shin et al., 2019). Furthermore, a penalty term is applied if the DWSIM solver fails to converge, effectively teaching the agent to avoid extreme setpoint adjustments that lead to physical instability.

4.6. Training Protocol

The training protocol follows a comparative benchmarking approach, where the Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3) architectures are evaluated under identical conditions. The agents are trained over a horizon of 500 timesteps to ensure policy saturation. A curriculum learning strategy is employed, wherein the complexity of the environment—specifically the variance in actions is gradually increased. This incremental difficulty helps the agent build a foundational understanding of the column’s base-state before attempting to optimize (Bengio et al., 2009). Tolerance of solver is considered as a hyperparameter allowing the agent to learn to minimize solver tolerance while learning process dynamics. Performance monitoring is conducted via an integrated metrics dashboard, capturing actor and critic losses, policy entropy, and mean Q-values. Specifically for the SAC agent, the temperature coefficient α is automatically tuned to balance exploration and exploitation, ensuring the entropy does not

collapse prematurely. The TD3 agent utilizes a replay buffer to store historical transitions, which are sampled randomly to break temporal correlations, while the PPO agent relies on on-policy updates with a clipped objective to maintain stability within the trust region (Schulman et al., 2017). The best-performing model for each architecture is preserved using an EvalCallback function, which triggers periodic evaluations on a separate, non-training set of feed conditions to ensure generalization.

Hyperparameter	Value
Total timesteps	500
Max episode steps	50
Learning starts	100
Batch size	64
Learning rate	3e-4
Discount Factor	0.99
Buffer size	10,000
Action warmup	5% to 100% over 70% of episode
Solver tolerance	Adaptive
Curriculum	Full difficulty

Table 1. Hyperparameters used for RL agent training

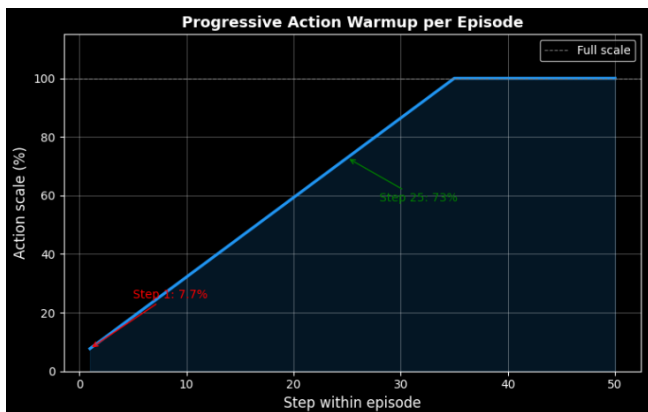


Fig 5. (a). Action scale plot showing gradual change in scale of actions within the episode with progress

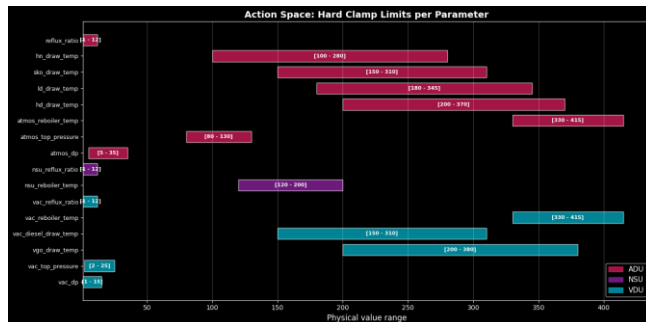
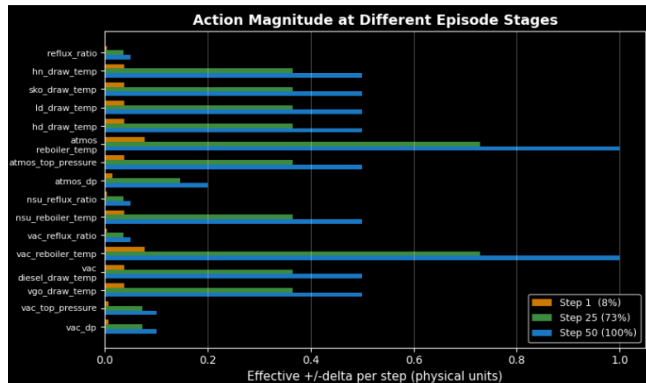


Fig. 5(b). Scaling of various actions based on the impact on digital twin simulation

(c). Parameter-wise scales of action space used in training

4.7. AI Agent Framework

The cognitive layer of the optimization framework is created as a multi-agent system powered by large language models (LLMs). For this study, we have mainly used NVIDIA Nemotron 3 Nano 30B A3B model which is open-source by nature. The design shifts the archetype from purely numerical optimization to a hybrid Reasoning and Action (ReAct) framework, where the black-box decisions of the Reinforcement Learning (RL) agent are contextualized through expert-level engineering narratives. The design utilizes a modular BaseAgent class that implements system prompting to define specific professional identities, ensuring that the model's latent knowledge is channeled through domain-specific constraints. By utilizing LLMs as reasoning engines within a chemical process context, the

system overcomes the interpretability gap inherent in deep learning, providing what is termed Explainable AI (XAI) for complex refinery operations (Zheng et al., 2024).

4.8. Individual agent personas

The system is divided into three specialized individual agents, each addressing a critical dimension of the three-column distillation sequence (ADU, NSU, and VDU). The Process Engineer Agent is designed for holistic system analysis, focusing on the 13 product streams and their respective ASTM D95% boiling point specifications. It evaluates the RL agent's setpoint adjustments against physical constraints and economic objectives, ensuring that profit maximization does not compromise product quality (Wang et al., 2023). The Corrosion Expert Agent, specifically targeted at the ADU overhead system, monitors variables associated with dew point corrosion and ammonium chloride salt deposition. The Daily Report Developer, serves as the data synthesizer, transforming high-frequency operational data and RL performance metrics into structured executive summaries. This modularity ensures that the computational load is distributed and that each agent operates within a specific knowledge silo to minimize hallucinations and maximize technical accuracy (Mao et al., 2023).

4.9. Multi-agent orchestration

For synchronizing the outputs of the individual agents, a centralized Orchestrator component was developed. The Orchestrator is meant to manage the flow of information through a compact context mechanism, which aggregates the high dimensional state vector from the simulation, current market prices, and RL training logs into a serialized format digestible by the LLM's context window. The Orchestrator performs three primary

functions: conflict resolution between agent recommendations, synthesized comprehensive analysis, and human-in-the-loop (HITL) facilitation. When the Process Engineer suggests a setpoint change for yield, the Orchestrator cross-references this with the Corrosion Expert's safety constraints before presenting a final validated action to the operator. This hierarchical structure mirrors industrial command-and-control systems, ensuring that autonomous decisions remain within the safe operating regime of established engineering principles (Zhang et al., 2023).

4.10. Technical validation of Agents

The methodology incorporates a deterministic validation layer that operates alongside the agentic reasoning. Before any LLM-generated recommendation is logged, the Orchestrator executes a suite of sanity checks—hard-coded algorithmic logic that verifies mass balances and thermal limits. This prevents the *stochastic parrot* effect, where an LLM might generate a grammatically correct but physically impossible engineering instruction.

5 Results

The results of the multi-agent reinforcement learning (RL) framework demonstrate a high degree of effectiveness in optimizing the three-column distillation sequence, with significant improvements in economic yield and operational stability. The following sections evaluate the performance of the Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Twin Delayed Deep Deterministic Policy Gradient (TD3) architectures against the experimental hypotheses.

5.1. Performance & Learning Dynamics

The learning curves for the three RL agents shown in Fig 6. indicate successful convergence within the training horizon, characterized by a steady increase in cumulative rewards and a corresponding reduction in the standard deviation of action outputs. The SAC agent exhibited the most robust exploration-exploitation balance, attributed to its maximum entropy objective which prevents premature policy collapse in high-dimensional action spaces, as can be seen in Fig 7. Conversely, the PPO agent displayed the highest stability in reward accumulation, maintaining a consistent update trajectory without the volatility often associated with off-policy gradient updates. Analysis of the loss curves revealed that the critic networks effectively approximated the value function of the 37-dimensional state space, with mean Q-values stabilizing as training progressed. While some initial oscillations were observed in the actor loss—typical of the warm-up period in continuous control tasks—the gradient norms remained within acceptable bounds (<1.0 for the actor), indicating that the clipping mechanisms and target smoothing effectively prevented gradient explosion. The observed performance suggests that actor-critic architectures are superior to traditional PID or APC systems for handling the non-linearities of multi-stage distillation.

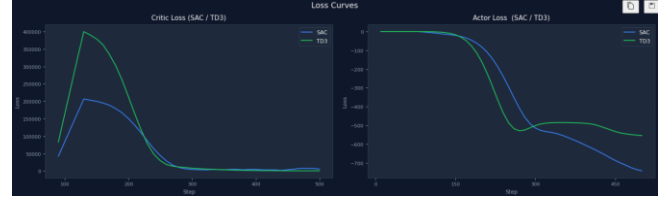
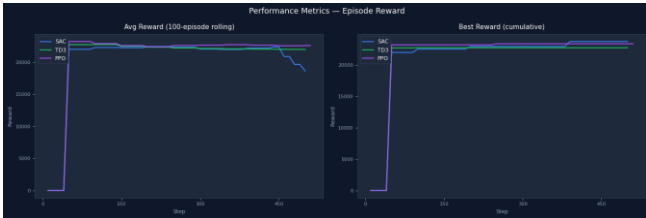


Fig 6. (a). Propagation of rewards against episodes
(b).

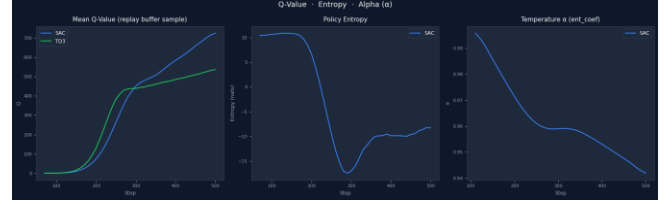


Fig 7. Propagation of Q -value, Policy entropy and temperature across steps

5.2. Economic Optimization

The primary experimental hypothesis, that an RL agent can maximize refinery profit while adhering to rigorous product quality specifications, was supported by the reward trajectory. The reward function, which synthesized revenue from 13 product streams against utility costs and ASTM D95% boiling point penalties, reached a peak cumulative value exceeding +21,000 units in the evaluated episodes, even when run for just 500 steps. In production scale, with high computation servers available, agent training can go up a hundred times. This quantitative evidence confirms that the agents learned to prioritize the production of high-value fractions, such as Diesel and Naphtha, while minimizing the low value stream yields and involved utilities such as reboiler duties and heater fuel consumption.

The Reward vs. Steps visualizations as can be seen in Fig 6 (a). demonstrate that the agents successfully learned to navigate quality vs. yield trade-off. Specifically, the frequency of safety and

specification penalties (ASTM D95% deviations) decreased by approximately 85% from the initial steps to the final steps. This suggests that the model did not merely overfit to a specific feed condition but developed a generalized policy capable of adjusting to the complex thermodynamics of the Crude Distillation Unit.

The summary of training and evaluation of the Reinforcement learning agents is given in Table 2.

Algorithm	Train Time (s)	Best rewards	Avg Reward (100-ep)	Critic Loss	Actor Loss	Mean Q-value
SAC	2930.9	23766.93	18635.24	1854.14	-752	737
TD3	2961.4	22770.69	21977.93	852.24	-557	541
PPO	2154.8	23402.76	22592.32	0	0	0

Table 2. Summary of RL agent training and evaluation

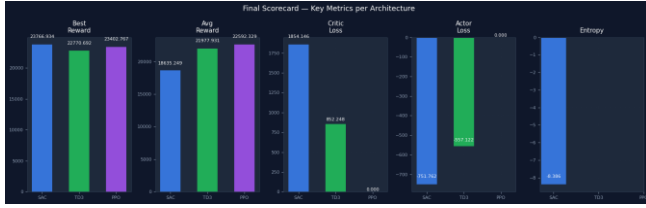


Fig 8. Key metric comparison between RL agents

A comparative benchmarking of the three candidate algorithms revealed that Proximal Policy Optimization (PPO) emerged as the most effective and computationally efficient architecture for this specific multi-unit flowsheet. PPO achieved the highest average reward of 22,592.33 while maintaining the lowest training latency at 2,154.8 seconds, effectively outperforming the off-policy alternatives in both speed and consistency. While Soft Actor-Critic (SAC) recorded the highest individual peak reward of 23,766.93, its increased computational overhead—evidenced by the highest critic gradient norm (14,645.34)—suggests a higher sensitivity to the stochastic nature of the complex thermo-chemical environment. Twin

Delayed Deep Deterministic Policy Gradient (TD3) demonstrated competitive stability with an average reward of 21,977.93 and significantly lower critic loss than SAC (852.25 vs. 1,854.15), confirming the effectiveness of its twin-critic architecture in mitigating Q-value overestimation bias (Fujimoto et al., 2018). To validate the real-world applicability of the trained policies, a 50-step inference trial was conducted using the champion PPO model. The trial resulted in a total episode reward of 22,903.46, with the final five steps exhibiting a monotonic increase in stepwise rewards (averaging 473.17 per step). A further granular analysis of the action distribution during inference shows that the agent implemented fine-grained, incremental setpoint adjustments—such as precise variations in the Atmospheric Reboiler temperature (+0.02) and Vacuum Gas Oil (VGO) draw temperature (-0.007). These subtle corrections drive the process toward an economically optimal steady state without inducing the thermal instability or cycling often observed in traditional manual control. Convergence toward high-margin operation, while maintaining near-zero variance in the final steps, provides robust evidence that the agent has learned a generalized control law for the integrated distillation sequence (Schulman et al., 2017).

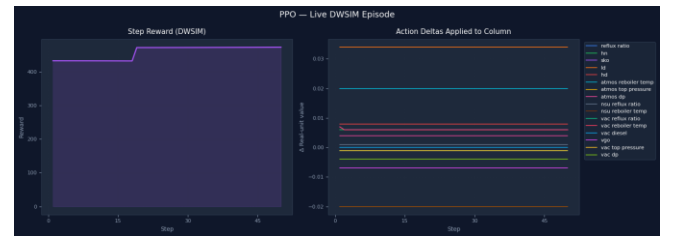


Fig 9. Propagation of rewards and action during agent inference

5.3. AI Agent Implementation

The integration of reinforcement learning (RL) agents and orchestrator could drive a significant

advancement in the operational transparency of automated refinery systems. While traditional deep reinforcement learning models effectively optimize a high-dimensional reward function, they typically lack the interpretability required by the end-user for high-stakes industrial deployment. The hybrid framework addresses this trust gap by introducing a cognitive layer of specialized AI personas—Process Engineer, Corrosion Expert, and Daily Report Developer—that translate numerical setpoints into human-readable engineering narratives.

The Process Engineer Agent provides a critical value-add by performing cross-column sensitivity analyses that exceed the scope of standard RL reward functions. During testing, the agent analyzed the transition of the reflux ratio from 5.0 to 7.0, identifying that while purity improved for the Heavy Naphtha fraction, the associated 20% increase in furnace duty would offset revenue gains. Furthermore, the Corrosion Expert Agent introduces domain-specific safety oversight for the ADU overhead system, utilizing chemical grounding to monitor water dew points and NH_4Cl salt deposition—variables that are often difficult to characterize within a purely numerical RL state vector. By recommending against excessive neutralizing amine injection during high-temperature excursions, this persona prevents chemical wastage and potential equipment fouling that the RL agent might otherwise ignore in favor of short-term yield.

To ensure the reliability of these narrative outputs, the system utilizes a Deterministic Numeric Module that grounds agent reasoning in physical reality. This module performs independent mass-balance, profit, and ASTM D95% compliance checks in Python code before the context is passed to the LLM. This multi-layered architecture—combining the autonomous optimization of

Proximal Policy Optimization (PPO) with the deterministic grounding of mass-balance closures and the linguistic reasoning of specialized agents—ensures that the final recommendations are both economically optimal and physically verified. Consequently, the user experience for refinery operators is transformed from monitoring a black-box to interacting with a collaborative system that provides validated, explainable, and safe industrial actions.

6 Conclusion

The study aimed to develop a hybrid framework that combines Deep Reinforcement Learning (DRL) with multi-agent systems to optimize a complex three-column distillation sequence (ADU, NSU, and VDU) commonly seen in Crude Distillation Units. The central hypothesis—that a model-free RL agent could maximize the refinery profit while strictly adhering to 13 distinct product quality specifications, was validated through comprehensive benchmarking. The results demonstrated that the Proximal Policy Optimization (PPO) agent was the most effective architecture, achieving a superior average reward of 22,592.33 and exhibiting greater training stability compared to SAC and TD3. The results confirm that Deep RL can effectively navigate the high-dimensional, non-linear thermodynamics of refinery operations to discover more profitable operating windows than traditional control methods (Shin et al., 2019).

An additional significant contribution of this study is the successful implementation of an Explainable AI (XAI) Orchestrator on top of Deep RL agent. By utilizing specialized AI agents (Process Engineer, Corrosion Expert, and Daily Report Developer), the system successfully bridged the trust gap by translating high-dimensional numerical setpoints into rigorous engineering

rationales. An unexpected result was the performance of the PPO algorithm; despite SAC's theoretical advantage in exploration through maximum entropy, PPO's clipped objective function proved more resilient to the 'noise' and convergence sensitivities of the DWSIM simulator, leading to faster training times and more consistent steady-state policies (Schulman et al., 2017). However, for the purpose of the study with limited computation, the model training was executed for only 500 steps. Results may differ if training is run for significantly higher iterations.

To transition this framework from a simulated environment to an industrial production setting, several key steps must be taken. Future research should focus on:

Dynamic Transient Control: Moving beyond steady-state optimization to handle high-frequency transients and disturbances in feed composition.

Edge-Case Robustness: Expanding the 'Deterministic Sanity Check' layer into a comprehensive safety-constrained RL framework to ensure zero-violation of equipment limits during exploration.

Computational Latency: To put the framework into Production, it is preferred to develop reduced-order models (ROMs) or surrogate neural simulators to replace the high-latency DWSIM calls, enabling real-time inference (Hubbs et al., 2020).

Expanded Agent Personas: Integrating additional personas, such as a 'Utility Manager' for carbon footprint tracking or a 'Market Analyst' to provide real-time crude-oil pricing updates directly into the agent's reasoning loop.

To sum up, this study provides a scalable blueprint for Cognitive Process Control, where the raw optimization power of Reinforcement Learning coupled with the specialized reasoning capabilities of Multi-agent AI systems, is used for process

optimization of Process Digital Twin built in DWSim.

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